



Employing large language models safely and effectively as a practicing neurosurgeon

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Abstract

Background Large Language Models (LLMs) have demonstrated significant capabilities to date in working with a neurosurgical knowledge-base and have the potential to enhance neurosurgical practice and education. However, their role in the clinical workspace is still being actively explored. As many neurosurgeons seek to incorporate this technology into their local practice environments, we explore pertinent questions about how to deploy these systems in a safe and efficacious manner.

Methods The authors performed a literature search of LLM studies in neurosurgery in the PubMed database (“LLM” and “neurosurgery”). Papers were reviewed for LLM use cases, considerations taken for selection of specific LLMs, and challenges encountered, including processing of private health information.

Results The authors provide a review of core principles underpinning model selection, including technical considerations such as model access, context windows, multimodality, retrieval-augmented generation, and benchmark performance, as well as relative advantages of current LLMs. Additionally, the authors discuss safety considerations and paths for institutional support in safe LLM inference on private health data. The resulting discussion forms a framework for key dimensions neurosurgeons employing LLMs should consider.

Conclusions LLMs present promising opportunities to advance neurosurgical practice, but their clinical adoption necessitates careful consideration of technical, ethical, and regulatory hurdles. By thoughtfully evaluating model selection, deployment approaches, and compliance requirements, neurosurgeons can leverage the benefits of LLMs while minimizing potential risks.

Keywords Large language models · Artificial intelligence · Neurosurgery

Abbreviations

LLM	Large language model	LLAMA	Large language model from Meta AI
AI	Artificial intelligence	RAG	Retrieval-augmented generation
ML	Machine learning	HER	Electronic health record
NLP	Natural language processing	HIPAA	Health Insurance Portability and Accountability Act
GPT	Generative pre-trained transformer	PHI	Protected Health Information

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Introduction

Large language models (LLMs) are a form of artificial intelligence that have recently garnered widespread popularity due to their ability to generate fluent human language in a versatile and robust manner. They have demonstrated highly accurate capabilities to date in working with a neurosurgical knowledge-base and have the potential to enhance neurosurgical practice and education [12, 20, 25, 29, 32, 34, 43]. However, their role in the clinical workspace is still being actively explored. As many neurosurgeons seek to incorporate this technology into their local practice environments,

questions remain on how to deploy these systems in a safe and efficacious manner. These are myriad, ranging from how to select an appropriate model for the given use case, to how to navigate local privacy and data safety regulations. Thus, while previous reviews have focused on LLMs in neurosurgery writ-large [9, 27, 35], to our knowledge no previous works have focused on surmounting the practical barriers towards deploying LLMs for the practicing neurosurgeon.

In this review, we seek to explore this question from a few different angles. First, we will review potential LLM use cases, discussing use cases both with and without incorporating patient protected health information (PHI). Second, we will summarize the technical components of deployment, including model selection and LLM-specific features to consider. Finally, we will consider broader patient and data safety topics and regulatory compliance.

Use cases: how might LLMs be employed by a neurosurgeon?

Use cases of LLMs in healthcare can be broadly divided into two categories by the nature of the data being processed. “Outside-the-firewall” applications do not involve the transmission of PHI, and thus are naturally compliant with local privacy regulations such as the United States Health Insurance Portability and Accountability Act (HIPAA). Application areas include literature review, medical education, patient education, and commonly banned uses such as for peer review. In contrast, “behind-the-firewall” uses like billing and coding, clinical document generation, report simplification, and data structuring are necessarily reliant on PHI, and thus subject to a greater degree of scrutiny. Here, we review progress and opportunities in each domain.

Use cases not involving PHI

Outside the firewall, LLMs have been employed successfully in simplifying patient-facing text. Doing so can improve readability, comprehension, and retention of material. Several authors have studied this use case on a variety of patient-facing materials. Mirza et al. found that the Kincaid readability score of consent forms decreased from that of a college student to that of a middle school student after simplification with ChatGPT, demonstrating how LLMs can be used to increase patient understanding in consent forms [20]. Similarly to Mirza et al., Swisher et al. demonstrated how LLMs can also be utilized to improve the readability of patient handouts and other patient-facing materials [20, 34]. Nian et al. evaluated ChatGPT’s response to frequently asked questions (FAQs) about spinal fusion and lumbar laminectomy. ChatGPT’s answers had a higher Flesch-Kincaid grade level score (11.6) compared to Google searches

(8.8) with longer responses [24]. These contradictory findings between studies highlight both the stochastic nature of LLMs and the importance of specific prompting towards desired output.

LLMs also hold significant potential in medical education. Not only have they demonstrated competent command of basic neurosurgical facts, but also have the potential to explain neurosurgical concepts to new learners, as well as generate a nearly infinite amount of practice questions to aid in understanding. Numerous benchmarking studies, including Gilson et al. demonstrated that LLMs like ChatGPT, can pass the United States Medical Licensing Exam (USMLE), functioning at the level of a third-year medical student [13]. The author further demonstrated how ChatGPT could help with learning as a virtual medical tutor. In a similar study, Kung et al. found that ChatGPT can perform at a level at or approaching the passing score of the USMLE exams, which include Step 1, Step 2 CK, and Step 3 [16]. Notably, this performance is achieved without further fine-tuning or further training on neurosurgery-specific text. This study demonstrated the improving accuracy of LLMs on USMLE exams and how LLMs such as ChatGPT can provide nuanced insights that may be useful in medical education.

One important limitation is the use of LLM in research and publication – whether that be in peer review or manuscript preparation. Currently, the US National Institutes of Health (NIH) expressly prohibits the use of LLMs in funding application peer review. Majovsky et al. found that there is potential for AI to create fraudulent nonscientific articles that are highly convincing.¹⁴ As Alkaissi et al. and many others highlight, the risk for hallucinations in LLM-generated text not subjected to human review is significant.¹⁵ Editorial attitudes towards using LLMs in manuscript writing are currently varied; some journals forbid it, many require a disclosure during submission, and some (such as NEJM AI) have taken a vocal stance to encourage it. Rapid changes in this technology also elicit changing judgements of the merits and shortcomings, and surgeons should be meticulous about reading individual organizational policies before employing LLMs in their scientific communication.

Use cases involving PHI

Incorporating PHI into LLMs significantly adds to their power and utility, but comes with additional regulatory scrutiny as well as privacy risks. While some organizations are working to adopt data-safe implementations of language models that are available for clinician use, efforts are still in early phases and are constrained by limited information technology staff and infrastructure. Nevertheless, many researchers have demonstrated strong use cases for incorporating PHI and LLM in several domains.

One area of active developing interest involves using language models to automate procedural billing. O'Malley et al. assessed the accuracy of various LLMs in looking up Current Procedural Terminology (CPT) codes for cranial, spine, and endovascular procedures using user-generated prompts, with reported accuracies ranging from 20 to 75% [25]. In a subsequent study, Roy et al. showed the capabilities of AtlasGPT and ChatGPT 4.0 in deriving CPT codes directly from operative notes in endovascular neurosurgery. They reported partial accuracies of 98.3% and 86.7%, albeit with complete accuracies of only 35.3% and 35.1%, respectively [28]. While current findings demonstrate the potential streamline efficiency, LLMs will inevitably need to be improved with further experience before broad deployment. They have also been proposed as a means to reviewing coding to reduce human error and it is purported that health insurers are already applying LLM in utilization management [40].

Another application focuses on processing clinical text, specifically in summarizing previously written documents or generating new notes. Van Veen et al. highlighted the use of GPT- 3.5 to synthesize key information from electronic health records into cohesive narratives, including summarizing imaging findings into impressions and creating problem lists from provider progress notes. Their findings indicated that a large percentage of these AI-generated summaries were either equivalent to (45%) or surpassed (36%) those generated by medical experts [41]. Dubinski et al. investigated the application of ChatGPT for creating discharge summaries and operative reports for neurosurgical patients, comparing its efficiency to standard speech recognition software [12]. They found that the median time for drafting discharge summaries for complication-free chronic subdural hematoma dropped from 16 min with standard software to just 1.85 min, achieving a median accuracy of 83% as evaluated by senior physicians. In another study, Ali et al. trained GPT- 4 to draft detailed operative notes using surgeon-specific templates and key parameter inputs, with independent evaluations by three blinded neurosurgeons on a scale of 1 to 5, showing no significant differences in accuracy between AI-generated notes (4.44) and those created by surgeons (4.33) [3]. These studies suggest that large language models (LLMs) have the potential to streamline clinical documentation, particularly for routine and uncomplicated procedures in neurosurgery.

Leveraging LLMs for mining and structuring patient data from electronic health records for research purposes is another area of interest. Truhn et al. demonstrated the ability of GPT- 4 to extract structured data from unstructured pathology reports for diffuse adult-type glioma [38]. They found that GPT- 4 achieved an error rate of just $1.0 \pm 0.4\%$ in extracting the IDH mutation status, which underscores its precision and efficiency in data extraction tasks. This

allowed them to observe significant correlations between the Ki- 67 labeling index and the presence of necrosis, as well as increased Ki- 67 labeling when IDH wild-type was present. Similarly, Wang et al. explored a biomedical language model, BRLM, to classify genomic variants and construct networks linking clinical symptoms, biomarkers, and related diseases [42]. Their model successfully identified genetic mutations and used a cumulative perturbation measurement for pathways. The model's perturbation scores helped in understanding the impact of genetic mutations, potentially enhancing the precision of treatment strategies through clear insights into pathogenicity mechanisms. By structuring and analyzing vast amounts of patient data, these models can identify patterns and trends that might otherwise remain obscured, thereby supporting evidence-based practice and personalized medicine.

Model selection: how do I go about selecting a model?

Core principles

Choosing the appropriate LLM for a neurosurgical task necessitates a comprehensive evaluation of its accuracy, reliability, interpretability, and compliance with regulatory standards. In addition, the specific nature of the task—which includes parameters such as the amount of text being processed and the need to update a model's knowledge base regularly—are critical factors to consider. Here, we discuss the two main dichotomies of model types (open source versus closed source models) and survey technical parameters influencing model selection, performance on medical benchmarks, handling of data types, advanced LLM frameworks such as Retrieval Augmented Generation (RAG), and neurosurgery-specific benchmark performance. A summary of current leading LLMs and their strengths is given in Table 1.

Open source vs. closed source

LLMs can be broadly categorized as open source or closed source models, with each presenting unique advantages and challenges [46]. Open source models are defined as those whose weights are publicly available and can be run on local devices or servers. Prominent examples include Meta's Llama series of models and a large collection on Hugging Face (a private company which develops machine learning tools and hosts libraries of models, datasets on its platform) [18, 44]. In contrast, closed source models like OpenAI's ChatGPT and Google's Gemini do not release model parameters or weights to the public [37].

These two LLM paradigms come with unique hosting considerations, or in other words, where a model's computations

Table 1 Example LLM models and their relative advantages

Model family	Developer	Private or open source	Advantages
GPT- 4o	OpenAI	Private	High zero-shot performance across tasks
o1, o3	OpenAI	Private	Advanced reasoning capabilities
Gemini	Google	Private	1-million-token context window with Gemini 1.5 Flash
Deep research	OpenAI	Private	Multi-step tasks and information synthesis
Claude	Anthropic	Private	Strong performance with writing and coding assistance
Llama 2	Meta AI	Open Source	Focus on safety and factual accuracy
DeepSeek-R1	Deep Seek	Open Source	Open source model rivaling large private LLM performance
Phi	Microsoft	Open Source	Low-cost small LLMs

directly occur. Local hosting provides enhanced control of the data, making it a good option for applications involving sensitive patient data due to user-determined data security practices. However, only open source models can be hosted on local computing devices. In addition, local hosting often incurs significant upfront capital investment to develop adequate computing infrastructure (e.g., GPUs) to fulfil user needs [22].

Cloud-based solutions offer better scalability, essentially incurring costs for only what you use and having nearly no upper limit on use, and access to continuously updated model versions and incur reduced costs through payment on a per-token basis with inference being handled by a dedicated server. However, careful assessment of regulatory compliance before adoption is needed as most cloud-based computing resources do not comply with health data privacy guidelines. Options such as Microsoft Azure OpenAI Service can be used in a privacy compliant manner through Business Associate Agreements (BAAs) [33].

Context window

An LLM's context window refers to the amount of text that can be processed by the model in a given request. Modern long-context LLMs with capacities exceeding 100,000 tokens can enable comprehensive analysis of full patient histories without requiring fragmented input, thereby improving contextual continuity. A model with a limited token window may fail to recognize key longitudinal data trends, potentially impacting clinical decision-making. However, while models with longer context windows have greater capabilities, they also incur greater computational costs, which may manifest as greater resources needed or longer response times. Additionally, long-context models struggle with “needle in a haystack” problems where important information is buried in the middle of a large input sequence [17]. These factors present an essential tradeoff that should be navigated through determination of the minimum context length required to comfortably satisfy user queries.

Multimodality

Advancements in multimodal AI allow models to process both textual and visual data. LLMs with integrated vision capabilities, such as GPT- 4-Vision, can potentially interpret imaging studies in conjunction with clinical notes, facilitating enhanced diagnostic support and decision-making [7, 36, 47]. This capability holds promise in areas like neuroradiology, where automated AI-driven image-text synthesis can accelerate workflow efficiency. However, studies have shown that leading vision-language models (VLMs) disproportionately exploit text over imaging input when generating their output [8]. Additionally, performance of VLMs is often poorer on text-only tasks when compared to LLMs. Therefore, a suggested principle is to only use VLMs when interpretation of imaging or video data is necessary until performance equilibrates or improves.

Retrieval-augmented generation

One inherent limitation of LLMs is the difficulty associated with updating the core knowledge of an LLM as guidelines change, which involves either completely retraining or finetuning the LLM on updated datasets. This is also often challenging or impossible with closed-source models, which cannot be retrained by users, and is extremely computationally (and financially) expensive for open-source models. Retrieval-Augmented Generation (RAG) is a technique that retrieves relevant documents from a user-created database based on a user query and provides that to an LLM, increasing the accuracy and reliability of LLM output and grounding it in external facts (Fig. 1) [23]. A prominent closed-source example of a RAG-enabled LLM in neurosurgery is AtlasGPT, and a popular open-source framework to create custom RAG frameworks that are compatible with any LLM is LangChain [10, 15].

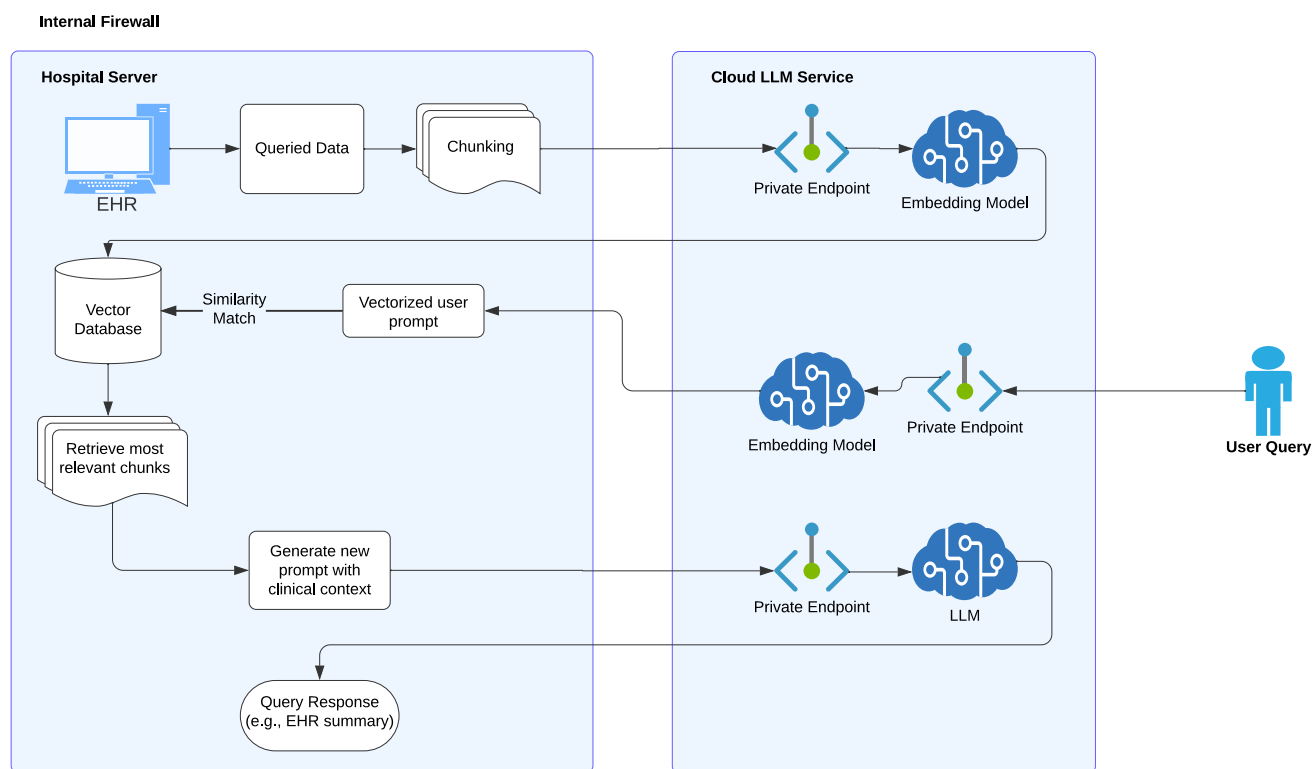


Fig. 1 Example workflow for a custom HIPAA-compliant retrieval-augmented generation LLM system in the cloud

Performance on neurosurgical benchmarks

Many studies have benchmarked the performance of LLMs on standardized neurosurgical questions [1, 2, 6, 14, 19, 21, 29, 31, 43]. Several studies have explored LLM performance on medical licensing exams, particularly in neurosurgery, with varied outcomes. Most investigations focused on U.S. licensing tests, often using the Self-Assessment Neurosurgery Examinations (SANS) from the American Board of Neurological Surgery. Ali et al. reported that GPT-4 outperformed both GPT-3.5 and Bard, surpassing the average human score, while all tested LLMs achieved a passing mark on the SANS question set [1]. Other research examined LLMs in international neurosurgical exams. Stengel et al. assessed multiple models on the European Board Examination in Neurological Surgery (EANS) and found that Bard had the highest accuracy at 62%, exceeding the human average of 59% [32]. Similarly, Sahin et al. evaluated Turkish neurosurgical licensing exams, noting that GPT-3.5 scored above the average candidate and provided clearer responses [29].

LLMs have struggled outside of conventional multiple-choice assessments, however. In an adapted Turing Test, Williams et al. had evaluators—who were unaware of the sources—assess interview transcripts from both ChatGPT and U.K. physicians applying for a neurosurgical National

Training Number [43]. Their findings revealed that ChatGPT underperformed compared to six of the eight human candidates, never surpassed the average score of those who secured training positions, and sometimes produced inaccurate responses that could be dangerous in a clinical setting.

Monitoring and regulatory compliance: how do we keep patients safe?

Privacy/data safety compliance

Policies around privacy and data safety compliance in the use of AI systems within healthcare are guided by a complex regulatory framework that varies across regions. In the United States, adherence to the mandates set by the Food and Drug Administration (FDA) and the Health Insurance Portability and Accountability Act (HIPAA) is crucial for safeguarding patient health data privacy. In December 2024, the U.S. Department of Health and Human Services (HHS) suggested updates to both the HIPAA Privacy Rule and the HIPAA Security Rule to bolster cybersecurity and patient data [26]. Many of these updates align with the General Data Protection Regulation (GDPR) in the European Union, which demands robust

protections for personal data, and therefore pertain to AI systems that handle patient information [5].

Compliance with these guidelines should be thought through carefully, as there are several nuances to be aware of. For instance, there is a vigorous debate on the distinction between "mandatory" and "strongly encouraged"—language that appears in numerous policies and dictates the level at which the policies should be implemented [11]. In addition, the GDPR emphasizes that patients should be informed about the usage of their health data in AI applications and should provide consent for any AI-driven analysis or decision-making. Data minimization is also prioritized, ensuring that only the essential data is collected for AI purposes. Critically, both the GDPR and HIPAA mandate a de-identification strategy to facilitate the safe application of AI, thus minimizing the risk of compromising patient privacy (European Parliamentary Research Service, 2020; U.S. Department of Health and Human Services, 2025). This de-identification process can be executed through either the Expert Determination method or the Safe Harbor method, involving the removal of specific identifiers (45 C.F.R. § 164.514(b)(2)).

Encryption is a key aspect of regulatory compliance. The GDPR regulations, particularly those outlined in Article 32, strongly advocate for the encryption of personal patient data both at rest and in transit. They encourage the adoption of advanced techniques like field-level, homomorphic, or tokenization to ensure secure data processing without the necessity of decrypting sensitive information [4]. Encrypting specific components of electronic health records can be particularly beneficial for researchers utilizing language models to access and analyze relevant patient information while ensuring the privacy of personal identifying information. Moreover, the Fairness of Artificial Intelligence Recommendations statement underscores the importance of incorporating fairness and transparency in AI healthcare applications. It supports the encryption of data that may pose a risk to patient privacy and promotes the implementation of best practices to guarantee equitable AI deployment [39].

To advance data safety and privacy compliance in AI healthcare applications, the existing regulatory frameworks must evolve to address current gaps and to anticipate future technological advancements. Although HIPAA provides foundational privacy protections, it currently lacks the comprehensive scope of the GDPR, indicating a need for reinforced legislation to close existing loopholes and ensure stronger data protection in the U.S. The challenges are mounting due to evolving uses and potential policies in this space. Additionally, a significant hurdle in the U.S. is the variation in state-by-state privacy policies for patient data, with many states lacking applicable laws.

Institutional-specific support and institutional policies

In our opinion, institutional-level support for LLM deployment is critical. Institutional-scale initiatives can ensure proper compliance, as well as provide a partner for testing, scaling, and even commercialization of LLM systems.

At one level, institutional support for basic technology infrastructure can be immensely helpful in dealing with PHI safely. Our own institution and many of its peers host HIPAA-compliant LLM instances available to all affiliated physicians and researchers that can interface with protected data for both research and non-research purposes. These are supported by in-house technical support staff, and are accessible by general program language (e.g., Python) application programming interfaces (APIs). In our experience, these institutionally sponsored instances are some of the best ways for neurosurgeons to transition to PHI-specific deployments, as many of the regulatory and compliance issues are standardized, freeing the individual surgeon to focus on the clinical implementation.

Coincident with this, ethical guidance through institutional review boards (IRBs) is necessary for any research applications. Not only is this a critical regulatory step for conducting research, but can also provide a valuable partners in helping to protect vulnerable populations that may be impacted by the work. Furthermore, IRBs can serve as a conduit to community leaders and group representatives if the research impacts populations that cannot practically be consented for use with LLM data (e.g., in trauma applications).

Increasingly, institutions are also important partners in intellectual property protection and commercialization of ideas. If successful, surgeon-led AI initiatives can be the basis for valuable intellectual property, and surgeons must consider how they want their idea to be protected and disseminated. Commercialization not only is a financial consideration for surgeons personally but can be an important way to promote and disseminate key discoveries. As part of this, surgeons' employment contracts should be reviewed carefully, and professional legal advice can be very helpful in understanding how any developed intellectual works relate to their employer. Many times, hospitals and universities will have a monetary interest in promoting the surgeon's work and have developed in-house "incubators" or "accelerators" to support commercialization, protection, and dissemination of surgeons' ideas.

Handling hallucinations and omissions

As of the writing of this article, current LLMs are still susceptible to producing statements that are either erroneous (i.e., "hallucinations") or missing critical requested

information (“omissions”). These can stem from a combination of idiosyncrasies in their original training data and the inherent probabilistic nature of LLMs. These pose significant problems for safe clinical deployment, as LLM prompt responses do not always come with proper confidence markers (i.e., the problem of being “confidently wrong”) and the potentially onerous nature of fact-checking the produced text, which may negate the utility of utilizing the LLM in the first place.

As such, it is paramount to design systems that can efficiently and effectively mitigate errors in any LLM-based system. Strategies for this fall into a few different categories. There are increasingly sophisticated technical solutions. These include anchoring responses in relevant “ground truth” text, via RAG-integration, as well as producing confidence measures along with text output – allowing for more targeted fact-checking [30, 45]. In addition, limiting the response length to only the information necessary to complete the task, and providing as much structure to LLM-fed prompts, are best practices that can limit the chance for errors and make fact-checking more efficient. There even have been several proposed methods whereby LLMs can try to fact-check themselves, which may hold near-term promise.

However, in clinical decision-making settings, where patient safety is reliant on LLM output, we strongly advocate for direct human oversight and accountability for all LLM-produced output. Much like the use of any other piece of technology in the operating room, surgeons are ultimately responsible for understanding the limits of their tools and for maximizing patient safety. Designing ways for clinicians to efficiently monitor LLM output is key to operational success.

Conclusions

LLMs offer exciting opportunities to enhance neurosurgical practice, but their clinical integration requires careful navigation of technical, ethical, and regulatory challenges. By systematically addressing model selection, deployment strategies, and compliance considerations, neurosurgeons can harness the potential of LLMs while mitigating associated risks. As the field continues to evolve, ongoing research and institutional collaboration will be essential to ensure the safe and effective use of this transformative technology in neurosurgical care.

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Declarations

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References

1. Ali R, Tang OY, Connolly ID et al (2023) Performance of ChatGPT and GPT-4 on neurosurgery written board examinations. *Neurosurgery* 93(6):1353–1365. <https://doi.org/10.1227/neu.0000000000002632>
2. Ali R, Tang OY, Connolly ID et al (2023) Performance of ChatGPT, GPT-4, and google bard on a neurosurgery oral boards preparation question bank. *Neurosurgery* 93(5):1090–1098. <https://doi.org/10.1227/neu.0000000000002551>
3. Ali A, Kumar RP, Polavarapu H et al (2024) Bridging the gap: can large language models match human expertise in writing neurosurgical operative notes? *World Neurosurg* 192:e34–e41. <https://doi.org/10.1016/j.wneu.2024.08.062>
4. Almalawi A, Khan AI, Alsolami F, Abushark YB, Alfakeeh AS (2023) Managing security of healthcare data for a modern healthcare system. *Sensors* 23(7):3612. <https://doi.org/10.3390/s23073612>
5. Artificial Intelligence Act: MEPs adopt landmark law | News | European Parliament. March 13, 2024. <https://www.europarl.europa.eu/news/en/press-room/20240308IPR19015/artificial-intelligence-act-meps-adopt-landmark-law>. Accessed 9 Mar 2025
6. Bartoli A, May AT, Al-Awadhi A, Schaller K (2023) Probing artificial intelligence in neurosurgical training: ChatGPT takes a neurosurgical residents written exam. *Brain Spine* 4:102715. <https://doi.org/10.1016/j.bas.2023.102715>
7. Blankemeier L, Cohen JP, Kumar A et al Merlin: a vision language foundation model for 3D computed tomography. *Res Sq*. Published online June 28, 2024:rs.3.rs-4546309. <https://doi.org/10.21203/rs.3.rs-4546309/v1>
8. Buckley T, Diao JA, Rajpurkar P, Rodman A, Manrai AK Multimodal foundation models exploit text to make medical image predictions. Published online November 25, 2024. <https://doi.org/10.48550/arXiv.2311.05591>

9. Bydon M, Shin JH, Timmons SD, Potts EA, Chan AK (2023) Introduction. Machine learning in neurosurgery: transitioning to a new era of contemporary medicine. *Neurosurg Focus* 54(6):E1. <https://doi.org/10.3171/2023.3.FOCUS23210>
10. Chase H LangChain. Published online October 2022. <https://github.com/langchain-ai/langchain>. Accessed 9 Mar 2025
11. Derisking AI: risk management in AI development | McKinsey. <https://www.mckinsey.com/capabilities/quantumblack/our-insights/derisking-ai-by-design-how-to-build-risk-management-into-ai-development>. Accessed 9 Mar 2025
12. Dubinski D, Won SY, Trnovec S et al (2024) Leveraging artificial intelligence in neurosurgery—unveiling ChatGPT for neurosurgical discharge summaries and operative reports. *Acta Neurochir (Wien)* 166(1):38. <https://doi.org/10.1007/s00701-024-05908-3>
13. Gilson A, Safranek CW, Huang T et al (2023) How does ChatGPT perform on the United States Medical Licensing Examination (USMLE)? The implications of large language models for medical education and knowledge assessment. *JMIR Med Educ* 9:e45312. <https://doi.org/10.2196/45312>
14. Guerra GA, Hofmann H, Sobhani S et al (2023) GPT-4 artificial intelligence model outperforms ChatGPT, medical students, and neurosurgery residents on neurosurgery written board-like questions. *World Neurosurg* 179:e160–e165. <https://doi.org/10.1016/j.wneu.2023.08.042>
15. Hopkins BS, Carter B, Lord J, Rutka JT, Cohen-Gadol AA Editorial. AtlasGPT: dawn of a new era in neurosurgery for intelligent care augmentation, operative planning, and performance. Published online February 27, 2024. <https://doi.org/10.3171/2024.2.JNS232997>
16. Kung TH, Cheatham M, Medenilla A et al (2023) Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models. *PLOS Digit Health* 2(2):e0000198. <https://doi.org/10.1371/journal.pdig.0000198>
17. Li T, Zhang G, Do QD, Yue X, Chen W Long-context LLMs struggle with long in-context learning. Published online June 12, 2024. <https://doi.org/10.48550/arXiv.2404.02060>
18. Llama. Meta Llama. <https://www.llama.com/>. Accessed 9 Mar 2025
19. Mannam SS, Subtirelu R, Chauhan D et al (2023) Large language model-based neurosurgical evaluation matrix: a novel scoring criteria to assess the efficacy of ChatGPT as an educational tool for neurosurgery board preparation. *World Neurosurg* 180:e765–e773. <https://doi.org/10.1016/j.wneu.2023.10.043>
20. Mirza FN, Tang OY, Connolly ID et al (2024) Using ChatGPT to facilitate truly informed medical consent. *NEJM AI* 1(2):AIcs2300145. <https://doi.org/10.1056/AIcs2300145>
21. Murphy Lonergan R, Curry J, Dhas K, Simmons BI Stratified evaluation of GPT's question answering in surgery reveals Artificial Intelligence (AI) knowledge gaps. *Cureus* 15(11):e48788. <https://doi.org/10.7759/cureus.48788>
22. Nagarajan R, Kondo M, Salas F et al (2024) Economics and equity of large language models: health care perspective. *J Med Internet Res* 26:e64226. <https://doi.org/10.2196/64226>
23. Ng KKY, Matsuba I, Zhang PC (2025) RAG in health care: a novel framework for improving communication and decision-making by addressing LLM limitations. *NEJM AI* 2(1):AIra2400380. <https://doi.org/10.1056/AIra2400380>
24. Nian PP, Saleet J, Magruder M et al (2024) ChatGPT as a source of patient information for lumbar spinal fusion and laminectomy: a comparative analysis against google web search. *Clin Spine Surg* 37(10):E394–E403. <https://doi.org/10.1097/BSD.0000000000001582>
25. O'Malley GR, Sarwar SA, Cassimatis ND et al (2024) Can publicly available artificial intelligence successfully identify current procedural terminology codes for common procedures in neurosurgery? *World Neurosurg* 183:e860–e870. <https://doi.org/10.1016/j.wneu.2024.01.043>
26. Office for Civil Rights. U.S. Department of Health and Human Services. "HIPAA Security Rule." HIPAA Security Rule NPRM. December 27, 2024. <https://www.hhs.gov/hipaa/for-professionals/security/hipaa-security-rule-nprm/index.html>. Accessed 9 Feb 2025
27. Patil A, Serrato P, Chisvo N, Arnaout O, See PA, Huang KT (2024) Large language models in neurosurgery: a systematic review and meta-analysis. *Acta Neurochir (Wien)* 166(1):475. <https://doi.org/10.1007/s00701-024-06372-9>
28. Roy JM, Self DM, Isch E et al (2025) Evaluating large language models for automated CPT code prediction in endovascular neurosurgery. *J Med Syst* 49(1):15. <https://doi.org/10.1007/s10916-025-02149-4>
29. Sahin MC, Sozer A, Kuzucu P et al (2024) Beyond human in neurosurgical exams: ChatGPT's success in the Turkish neurosurgical society proficiency board exams. *Comput Biol Med* 169:107807. <https://doi.org/10.1016/j.combiomed.2023.107807>
30. Savage T, Wang J, Gallo R et al (2025) Large language model uncertainty proxies: discrimination and calibration for medical diagnosis and treatment. *J Am Med Inform Assoc JAMIA* 32(1):139–149. <https://doi.org/10.1093/jamia/ocae254>
31. Sevgi UT, Erol G, Doğruel Y, Sönmez OF, Tubbs RS, Güngör A (2023) The role of an open artificial intelligence platform in modern neurosurgical education: a preliminary study. *Neurosurg Rev* 46(1):86. <https://doi.org/10.1007/s10143-023-01998-2>
32. Stengel FC, Stienen MN, Ivanov M et al (2024) Can AI pass the written European Board Examination in Neurological Surgery? - Ethical and practical issues. *Brain Spine* 4:102765. <https://doi.org/10.1016/j.bas.2024.102765>
33. stevevi. HIPAA - Azure Compliance. April 6, 2023. <https://learn.microsoft.com/en-us/azure/compliance/offering/offering-hipaa-us>. Accessed 9 Mar 2025
34. Swisher AR, Wu AW, Liu GC, Lee MK, Carle TR, Tang DM (2024) Enhancing health literacy: evaluating the readability of patient handouts revised by ChatGPT's large language model. *Otolaryngol Neck Surg* 171(6):1751–1757. <https://doi.org/10.1002/ohn.927>
35. Tangsrivimol JA, Schonfeld E, Zhang M et al (2023) Artificial intelligence in neurosurgery: a state-of-the-art review from past to future. *Diagnostics* 13(14):2429. <https://doi.org/10.3390/diagnostics13142429>
36. Tanno R, Barrett DGT, Sellergren A et al (2025) Collaboration between clinicians and vision-language models in radiology report generation. *Nat Med* 31(2):599–608. <https://doi.org/10.1038/s41591-024-03302-1>
37. Team G, Anil R, Borgeaud S et al Gemini: a family of highly capable multimodal models. Published online June 17, 2024. <https://doi.org/10.48550/arXiv.2312.11805>
38. Truhn D, Loeffler CM, Müller-Franzes G et al (2024) Extracting structured information from unstructured histopathology reports using generative pre-trained transformer 4 (GPT -4). *J Pathol* 262(3):310–319. <https://doi.org/10.1002/path.6232>
39. Ueda D, Kakinuma T, Fujita S et al (2024) Fairness of artificial intelligence in healthcare: review and recommendations. *Jpn J Radiol* 42(1):3–15. <https://doi.org/10.1007/s11604-023-01474-3>
40. Use generative AI for utilization management | Cloud Architecture Center. Google Cloud. <https://cloud.google.com/architecture/use-generative-ai-utilization-management>. Accessed 13 Mar 2025
41. Van Veen D, Van Uden C, Blankemeier L et al (2024) Adapted large language models can outperform medical experts in clinical text summarization. *Nat Med* 30(4):1134–1142. <https://doi.org/10.1038/s41591-024-02855-5>
42. Wang Y, Zuo J, Duan C et al (2024) Large language models assisted multi-effect variants mining on cerebral cavernous

- malformation familial whole genome sequencing. *Comput Struct Biotechnol J* 23:843–858. <https://doi.org/10.1016/j.csbj.2024.01.014>
43. Williams SC, Starup-Hansen J, Funnell JP et al (2024) Can Chat-GPT outperform a neurosurgical trainee? A prospective comparative study. *Br J Neurosurg* 0(0):1–10. <https://doi.org/10.1080/02688697.2024.2308222>
 44. Wolf T, Debut L, Sanh V et al HuggingFace's transformers: state-of-the-art natural language processing. Published online July 14, 2020. <https://doi.org/10.48550/arXiv.1910.03771>
 45. Zakka C, Shad R, Chaurasia A et al (2024) Almanac — retrieval-augmented language models for clinical medicine. *NEJM AI* 1(2). <https://doi.org/10.1056/aioa2300068>
 46. Zhang G, Jin Q, Zhou Y et al (2024) Closing the gap between open source and commercial large language models for medical evidence summarization. *NPJ Digit Med* 7(1):239. <https://doi.org/10.1038/s41746-024-01239-w>
 47. Zhang K, Zhou R, Adhikarla E et al (2024) A generalist vision-language foundation model for diverse biomedical tasks. *Nat Med* 30(11):3129–3141. <https://doi.org/10.1038/s41591-024-03185-2>

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